

Lei Shen

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EDUCATION

- **Wuhan University**, Wuhan, Hubei, China Sep 2023 – Present
M.S. in Medicine in Surgery, GPA: 3.71/4.0
Honors: National Scholarship for Graduate Students for two years (2023 & 2024)
- **Wuhan University**, Wuhan, Hubei, China Sep 2018 – May 2023
B.S. in Clinical Medicine, GPA: 3.58/4.0
Honors: Outstanding Graduate of Wuhan University (2023), Lei Jun Scholarship of Wuhan University (2022)

RELEVANT COURSEWORKS

- **Graduate-level:** Evidence-Based Clinical Medicine (A), Introduction to Clinical Research (A), Guided Reading of Medical Literature (A), Academic Ethics and Regulations (A), Surgery (B), Medical Professionalism (A), etc.
- **Undergraduate-level:** Basic Medicine Comprehensive Experiment (A), Clinical Pathophysiology and Therapeutics (A), Neuroscience (B), Surgical Internship (A), Internal Medicine Internship (A), etc.

PUBLICATIONS

- [1] Lei Shen*, Zhourui Zhang, Pengfei Wu, et al. "Mechanistic insight into glioma through spatially multi-dimensional proteomics". In: *Science Advances* 10.7 (2024), eadk1721. doi: 10.1126/sciadv.adk1721.
- [2] Jingyi Yang, Lei Shen*, Yuankun Cai, et al. "The Role of Coagulation-Related Genes in Glioblastoma: A Comprehensive Analysis of the Tumor Microenvironment, Prognosis, and Treatment". In: *Biochemical Genetics* (Mar. 2025). doi: 10.1007/s10528-025-11086-3.
- [3] Jingyi Yang, Lei Shen*, Jiabin Zhou, et al. "A Novel Mitochondrial-Related Gene Signature for the Prediction of Prognosis and Therapeutic Efficacy in Lower-Grade Glioma". In: *Biochemical Genetics* (Oct. 2024). doi: 10.1007/s10528-024-10928-w.
- [4] Lei Shen* and Nan Xiong. "Letter: Evaluation of 2 Surgical Techniques—Transposition Versus Interposition Microvascular Decompression for Hemifacial Spasm: A Systematic Review of 19 437 Patients". In: *Neurosurgery* (May 2025). Epub ahead of print. doi: 10.1227/neu.0000000000003496.
- [5] Lei Shen*, Mengyang Wang, Jingwei Zhao, et al. "Study on the relationship between obesity and complications of Pediatric Epilepsy surgery". In: *BMC Pediatrics* 23.1 (Mar. 2023), p. 142. doi: 10.1186/s12887-023-03948-9.
- [6] Lei Shen*, Jingyi Yang, Runqi Cheng, et al. "Bridge-layered decompression technique for vertebral artery-involved hemifacial spasm: technical note". In: *BMC Surgery* 24.1 (May 2024), p. 154. doi: 10.1186/s12893-024-02415-1.
- [7] Keyu Chen, Lei Shen*, Jingyi Yang, et al. "A nomogram based on clinical multivariate factors predicts delayed cure after microvascular decompression for hemifacial spasm". In: *Neurosurgical Review* 47.1 (Jan. 2024), p. 44. doi: 10.1007/s10143-024-02284-5.
- [8] Lei Shen*, Huan Liu, Xue Liu, et al. "An adolescent patient with anti-N-methyl-D-aspartate receptor encephalitis with motor aphasia as the first symptom and complicated by peripheral nerve damage: A case report and literature review". In: *Medicine (Baltimore)* 104.19 (May 2025), e42436. doi: 10.1097/MD.00000000000042436.
- [9] Jianmei Yang, Lei Shen*, Jingyi Yang, et al. "Complement and coagulation cascades are associated with prognosis and the immune microenvironment of lower-grade glioma". In: *Translational Cancer Research* 13.1 (Jan. 2024), pp. 112–136. doi: 10.21037/tcr-23-906.

SELECTED PROJECTS

Project 1: Functional mechanism of PEPD in glioblastoma (Master thesis)

Mar 2024 - Present

Location: Wuhan University, Xiong's Lab; Advisor: Prof. Nanxiang Xiong

Building on previous proteomic discoveries, my Master's thesis focused on PEPD as a key driver of glioblastoma (GBM) malignancy. I demonstrated that PEPD promotes GBM progression by inducing ZFP36-mediated alternative splicing of Insig1. This PEPD-driven splicing event disrupts the Insig1-mediated inhibition of SCAP/SREBP2-dependent cholesterol metabolism, thereby ultimately accelerating GBM malignant progression.

- Independently reviewed literature, formulated hypotheses, designed and executed research projects.
- Validated the role of PEPD–ZFP36–Insig1 axis in GBM using RNA-seq, qPCR, Western blot, Co-IP, and RIP.

Project 2: Spatially multidimensional proteomics of glioma

Sep 2021 - Mar 2024

Location: Wuhan University, Xiong's Lab; Advisor: Prof. Nanxiang Xiong

As a core member, I contributed to a multi-cohort proteomic study that uncovered novel molecular mechanisms in diffuse glioma. This collaborative work was published in *Science Advances* as the first-author paper titled "Mechanistic insight into glioma through spatially multidimensional proteomics".

- Performed proteomic sample preparation and downstream data analysis.
- Contributed substantially to data visualization and manuscript writing.

Project 3: The role of Cuproptosis in glioma progression

Mar 2022 - Jun 2023

Location: Wuhan University, Xiong's Lab; Advisor: Prof. Nanxiang Xiong

Cuproptosis is a novel type of programmed cell death, discovered in 2022. I led a *National Training Program of Innovation and Entrepreneurship* to investigate whether cuproptosis regulates glioma progression. Our findings revealed that activation of cuproptosis suppressed glioma development, suggesting a potential therapeutic value.

- Applied bioinformatics pipelines to analyze glioma-related datasets for cuproptosis signatures.
- Conducted *in vitro* cellular and molecular experiments to validate functional roles of cuproptosis.

RELEVANT TECHNICAL SKILLS

Wet-lab skills:

- Cell culture (staining, scratch assay, colony formation, CCK8, Transwell).
- Molecular techniques (qPCR, Western blot, Immunohistochemistry, ELISA, Co-IP, ChIP, RIP).
- Basic animal handling, sampling, and behavioral testing.

Dry-lab skills: R, Python, bioinformatic analysis for bulk and single-cell RNA-seq, proteomics, metabolomics.

Clinical skills: Lumbar puncture, continuous lumbar subarachnoid drainage, and basic surgical skills.

Languages: Chinese (native), English (proficient)

LEADERSHIP & COMMUNITY SERVICE

Academic Peer Review Service

Apr 2024 - Present

Serve as a reviewer for the *Heliyon* journal. Reviewed 2 papers.

Team Leader, Undergraduate Innovation Projects

Sep 2022 - Sep 2024

Led 6 undergraduate students to complete two *National Training Program of Innovation and Entrepreneurship* for Undergraduates, providing mentorship on research design, data analysis, and project execution.

Student Committee Member, School Student Council

Sep 2022 - June 2023

Coordinated with faculty to organize summer social practices, school sports events, and other student activities, ensuring smooth execution and student engagement.

Volunteer Coordinator, University Student Volunteer Association

Sep 2019 - Sep 2021

Organized and led 4 volunteer initiatives, including HIV/AIDS prevention campaigns and first-aid education programs, engaging 400+ students and 2 local communities.

Class President, Clinical Medicine Undergraduate Program

Sep 2018 - Jun 2020

Led a class of 23 students, organized academic and social activities to promote collaboration and cohesion among classmates, and coordinated with faculty on class matters.